

Problem Description:
Tina has given a boolean matrix `mat[P][Q]` of size `P X Q` to Laasya,
She wants to modify it such that if a matrix cell `mat[m][n]` is 1 (or true) then make all the cells of `m`_th row and `n`th column as 1. Can you help Laasya?

Constraints:
1 <= p, q <= 1000

Input Format:
First line of the input is the how many rows and columns in Laasya matrix.
After that, **each line will** represent each row and Laasya need to enter numbers.

Output Format:
Print the resultant matrix output in a separate lines.

Logical Test Cases

Test Case 1

INPUT (STDIN)

EXPECTED OUTPUT

2 2
1 0
0 1

1 1
1 1

Test Case 2

INPUT (STDIN)

EXPECTED OUTPUT

3 3
0 0 0
0 1 0
0 0 1

0 1 1
1 1 1
1 1 1

Mandatory Test Cases

Test Case 1

KEYWORD

int arr[r][c];

Test Case 2

KEYWORD

arrTemp[r][c]

Test Case 3

KEYWORD

for (m=0;m<r;m++)

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

13

Test Case 2

TOKEN COUNT

325

Test Case 3

NLOC

51

Code:

```
1  #include <stdio.h>
2  int main()
3  {
4      int p, q;
5      scanf("%d %d", &p, &q);
6      int matrix[p][q];
7      int row[p];
8      int col[q];
9      int i, j;
10
11     /* Initially no row or column is marked */
12     for (i = 0; i < p; i++)
13     {
14         row[i] = 0;
15     }
16
17     for (j = 0; j < q; j++)
18     {
19         col[j] = 0;
20     }
21
22     /* Read matrix */
23     for (i = 0; i < p; i++)
24     {
25         for (j = 0; j < q; j++)
26         {
27             scanf("%d", &matrix[i][j]);
28
29             if (matrix[i][j] == 1)
30             {
31                 row[i] = 1;
32                 col[j] = 1;
33             }
34         }
35     }
36     /* Make marked rows and columns 1 */
```

```
37     for (i = 0; i < p; i++)
38     {
39         for (j = 0; j < q; j++)
40         {
41             if (row[i] == 1 || col[j] == 1)
42             {
43                 matrix[i][j] = 1;
44             }
45         }
46     }
47
48     /* Print final matrix */
49     for (i = 0; i < p; i++)
50     {
51         for (j = 0; j < q; j++)
52         {
53             printf("%d", matrix[i][j]);
54
55             if (j < q - 1)
56                 printf(" ");
57         }
58
59         printf("\n");
60     }
61
62     return 0;
63 }
```

Output:

```
4 4
0 0 0 0
0 0 1 0
0 0 0 0
0 0 0 0
0 0 1 0
1 1 1 1
0 0 1 0
0 0 1 0
```

```
Process exited after 64.84 seconds with return value 0
Press any key to continue . . . |
```